



DUNSTAN



AUGUSTA UNIVERSITY



Develop a Sensor with Flood Detection Capabilities

Continuous Water-Level Sensor Based on Load Cell and Floating Pipe

research done by

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- Cost optimized
- Low power
- Environmentally friendly
- Remote monitoring



Archimedes Principle

F_B = Weight Displaced

$$= \rho_w g V$$

$$F_B = \rho_w F_g (\pi r^2 \cdot d)$$

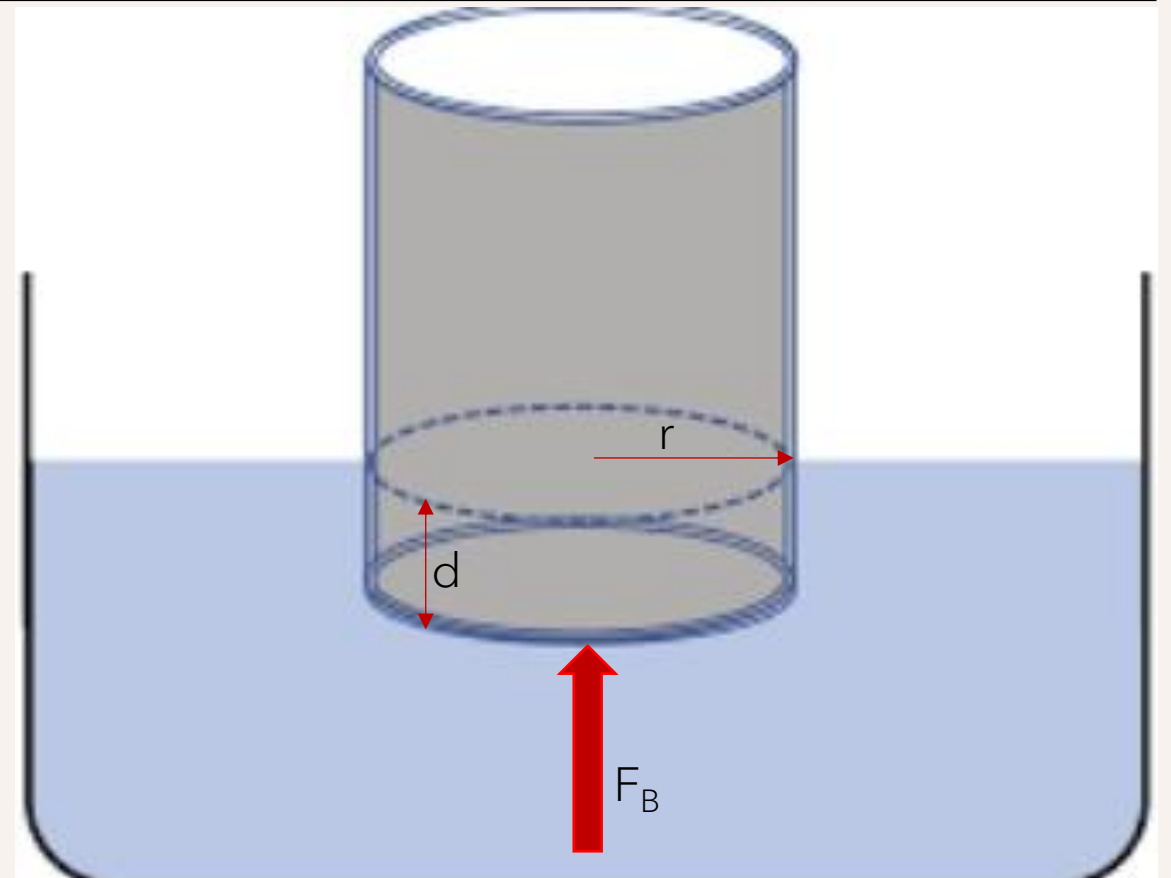
F_B = Buoyant Force

ρ = Density of water

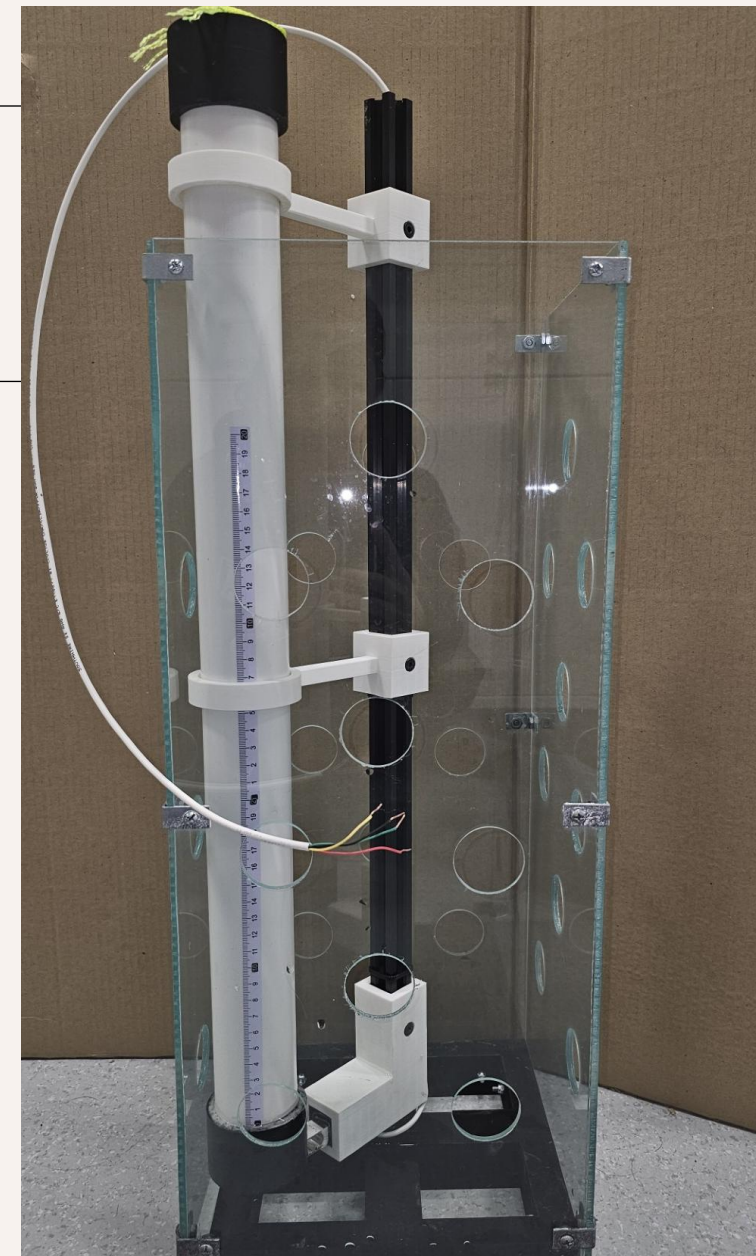
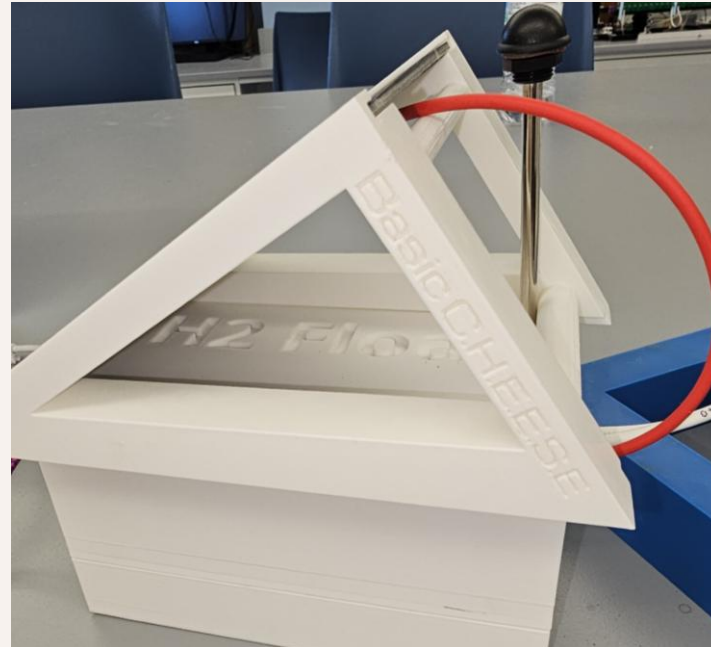
g = Gravitational acceleration

V = Volume of displaced water

d = Submerged Depth

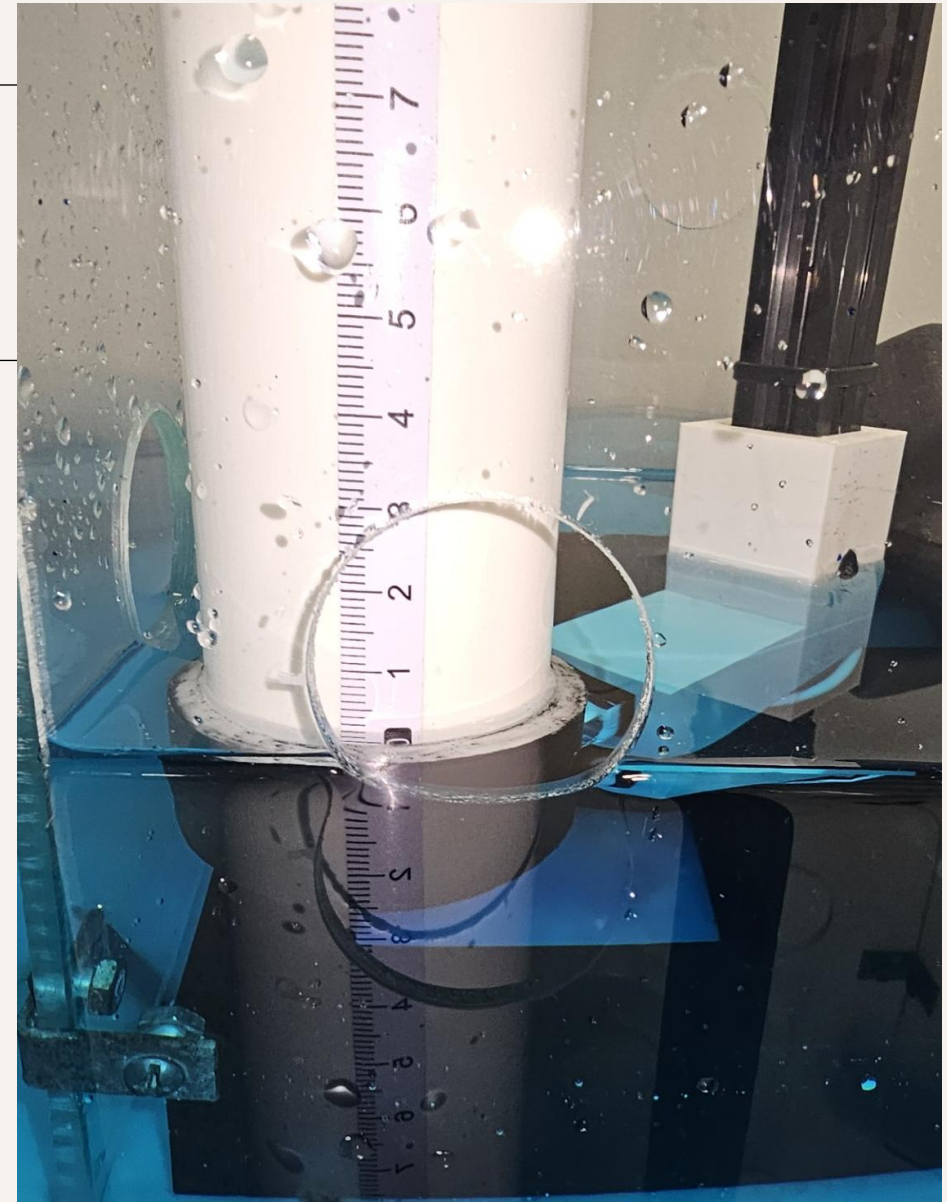


H₂Float Design 2.0

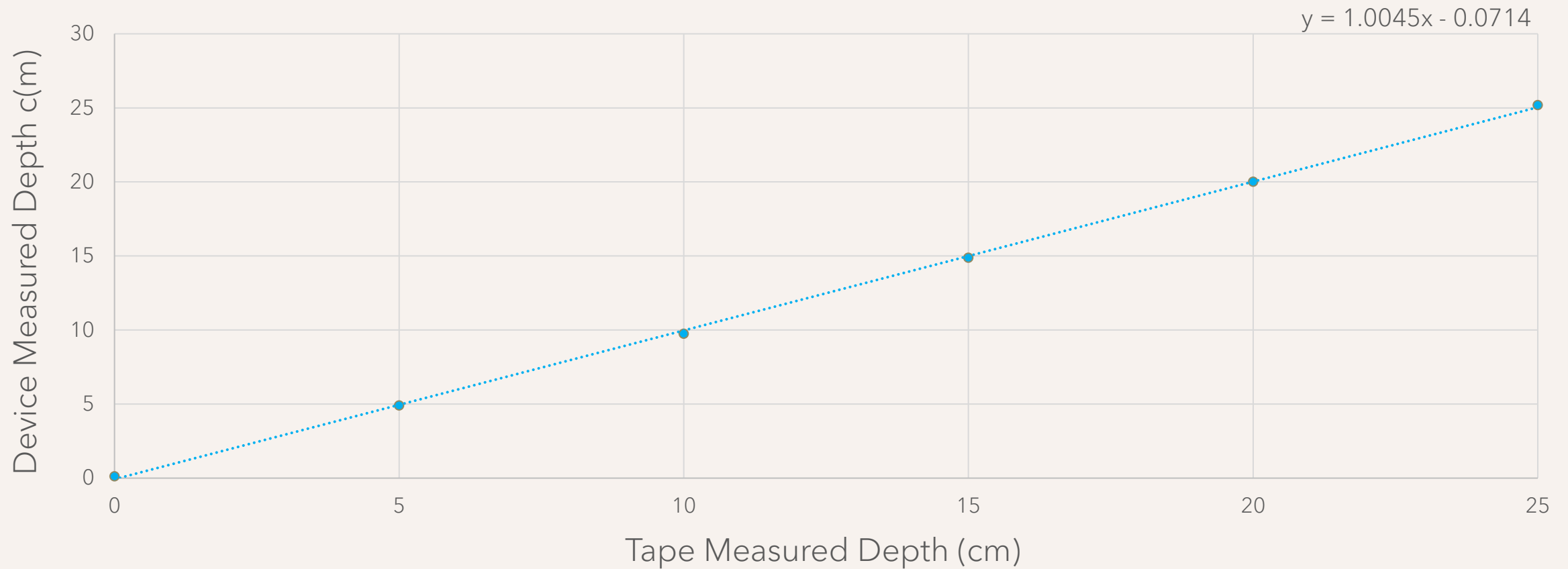


Progress Since Last Meeting

- Completed redesign and build of H₂Float 2.0
- Calibrated new build
- Measured depth while varying temperature



Measured Depth vs Tape Measured Depth



Measured Depth vs Water Temperature

